

UT INSPECTION REPORT TEMPLATE

Minimal Version Ultrasonic Thickness Survey — Final Inspection Report

This template defines the standard sections and structure for a UT thickness survey final inspection report. Each section contains a brief description of the content that should be populated from inspection data. For detailed field-level guidance, refer to the Detailed Version of this template.

Cover page fields:

- Report Number
- Revision
- Date
- Client
- Facility / Site
- Unit / Area
- Work Order
- Scope Summary
- Survey Dates
- Prepared By (Name, Certification, Date)
- Reviewed By (Name, Certification, Date)
- Approved By (Name, Certification, Date)

1. Executive Summary

Provide a high-level overview of the inspection: what was inspected, when, how many CMLs were surveyed, and the key outcomes. Summarize the count of findings by category (CRITICAL, Alert, Monitor, Acceptable, Inaccessible). Highlight the most significant findings that require immediate attention.

2. Scope of Work

Define what was inspected: list all piping lines/circuits with their line numbers, service type, pipe size (NPS), material, schedule, and CML count per line. Include the total CML count, survey date range, inspection method, and reference to the client's request for quotation or work order.

3. Reference Documents

List all applicable codes, standards, and procedures referenced during the inspection. Typical references include API 570, API 574, ASME B31.3, ASME Section V Article 5, API 579-1/ASME FFS-1, ASNT SNT-TC-1A, and any client-specific NDE procedures.

4. Personnel & Qualifications

List all inspection personnel with their names, roles, certification levels (e.g., UT Level II, UT Level III), certification numbers, employer, and confirmation that site-specific safety orientation was completed. Certification records should be appended.

5. Equipment & Calibration

Identify all UT equipment used: gauge model and serial number, transducer type and ID, calibration block ID, and material velocity settings. Include the daily calibration verification log showing pre-job and post-job readings against known block thicknesses with pass/fail status within the specified tolerance (typically ± 0.002 in). Reference appended calibration certificates.

6. Inspection Results — Data Tables

Present thickness measurement data organized by piping line and CML. For each CML, include: CML ID, location description, component type, nominal wall thickness, minimum required thickness (t-min), previous survey thickness, current minimum thickness, and disposition category. Also include the individual quadrant readings (typically 4 readings at 90° intervals), measurement date/time, operator initials, and gauge serial number.

7. Trending & Corrosion Analysis

Calculate and present corrosion rates and remaining life for each CML. Include short-term corrosion rate (based on previous survey), long-term corrosion rate (based on baseline/install), remaining life in years, and recommended next inspection date. State the formulas used and the basis per API 570. Flag any anomalous conditions (e.g., readings exceeding nominal, suggesting scale deposits).

8. Findings & Dispositions

Categorize all CML findings using the standard disposition categories: CRITICAL (below t-min), Alert (below 110% of t-min), Monitor (corrosion rate exceeding threshold, typically >10 mpy), and Acceptable. For each non-acceptable finding, state the basis for the classification and the required disposition or action (e.g., FFS evaluation, repair, increased monitoring).

9. Recommendations

Provide recommended actions for each finding category: recommended re-inspection intervals (typically half of remaining life, bounded between 6 months and 5 years), referrals for engineering review or FFS evaluation, process review recommendations, and any additional CML locations to add for future surveys.

10. Appendices

Include the following supporting records as appendices:

Appendix A: Raw Data (Gauge Data Exports)

Raw thickness data exported from the UT gauge in CSV or native format.

Appendix B: Photograph Log

Photos of each CML location (wide shot + close-up), gauge display photos for CRITICAL and Alert readings, surface condition photos, and inaccessible location photos.

Appendix C: Personnel Certification Records

Copies of ASNT SNT-TC-1A or equivalent UT certification for all inspection personnel.

Appendix D: Equipment Calibration Certificates

NIST-traceable calibration certificates for the UT gauge, transducer, and calibration block.

Appendix E: Isometric Drawings with CML Locations

Marked-up isometric drawings showing the physical location of each CML on the piping.